Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/25/2006
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Project Title:	Advanced Materials Growth in Microgravity: Carbon Nanotubes, Semiconductors and Advanced Nanocomposites		
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Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
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Space Biology Special Category:	None		
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Zip Code:	75083-0688	**Congressional District:**	3
Comments:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Macknight, A. (Honeywell Space Systems (Torrance) Baughman, Ray (UTD-NanoTech Institute)		
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	The major task to build the arc-discharge chamber in which the geometry of electrodes and the gas flows will create the conditions, simulating microgravity is presently finished. We have successfully checked the influence of vertical versus horizontal position of electrodes and found the conditions for optimal growth of best choice SWCNT. This data can be now used to create a new geometry of electrodes, which together with optimal gas flow results in the semi-continuous growth of SWCNT. We found, that nanotube bundles grow thicker and longer than in conventional conditions, due to longer presence in the reaction zone. We have checked the hypothesis to rotate one electrode and thus to obtain the torroidal distribution of reaction zone in the distributed arc. However we were not able to obtain the selective separation of tubes. We have started to build an advanced CVD reactors for the synthesis of tubes of exotic shapes and geometries, taking advantage of simulated microgravity conditions.		

Task Description:	The purpose of the program is: To simulate microgravity synthesis of carbon nanotubes (CNT) using new non-solvent based separation technology, magnetic fields for flow control, and high-efficiency arc discharge, spark discharge and chemical vapor deposition techniques. NASA has been developing new concepts in materials science and molecular nanotechnology based on single-wall carbon nanotubes for revolutionary electronics concepts, and other materials applications. A new innovative Chemical Vapour Deposition chamber (CVD) method for synthesis in simulated microgravity will hopefully make single-wall nanotubes (SWNT) in a confined circulated vapour geometry, which we expect will be similar to zero-g conditions and will enable manufacture of CNTs in larger quantities. Theoretically such long SWNTs should have superior theoretical strength, high thermal conductivity, exceptional thermal stability, and numerous potential applications. Among the novel applications that have been considered for SWNTs are nanomachine technology (nanotube-based gears); carbon nanotube electronic components such as supercapacitors, battery electrodes, actuators; and hydrogen storage devices for fuel cells, space vehicles, and other applications where high-density hydrogen storage is required. Carbon nanotube materials grown in a microgravity environment in our new CVD device (shown at Fig 1, a) and b) will have higher strength and superior properties due to larger defect-free structures. In order to ensure the high quality of these materials, a containerless method of growing them on a flying catalyst (Ni or co nanoparticles) must be used wherein they do not contact the walls of the reactor, thus ensuring better control of the crystallization process. These containerless conditions can be realized easily in microgravity flight experiments, and will be first simulated in ground testing CVD with the help of external magnetic and microwave fields which provide temperature and location control. Longer and stronger Carbon Nanotubes, created using these techniques will have a tremendous impact on earth and space technologies of the near future. As the result of the first year of the program we have obtained CNTs by advanced arc-discharge synthesis with larger output in so called reversed geometry of electrodes in which a role of convective flow is enhanced, as in microgravity conditions. We also have built a basic CVD device, in which during last year of the project we will try to grow carbon nanotubes of unusual geometries, and particularly the spiral ones. First objective is to further develop the arc-discharge set-up which simulates microgravity conditions, and allows to use the advantages of microgravity for the growth of improved single wall carbon nanotubes (SWCNT): longer tubes with smaller number of defects. Second objective is to use this experimental set-up for increasing the quantity and efficiency of SWCNT production and achieve selective separation of SWCNTs (metallic from semiconducting, and armchair ones from chiral tubes) and to create advanced nanocomposites of SWCNTs with semiconductors. One important part is the computer simulation of the growth of SWCNTs and semiconductors in a contact-less geometry in which the nanocomposite can be formed. This task has been finalized this year. Usual arc-synthesis or CVD growth of SWCNT results in rather short (up to 10 microns) nanotubes, because the time of reaction of carbon atoms in arc-plasma with catalytic particles of Co or Ni/Y clusters is restricted. Moreover the SWCNTs are obtained on the walls of the reactor as a mixture of bundled metallic and semiconducting tubes. Therefore the ability to have a controlled prolonged growth in a plasma continuously levitating in a microgravity would be of major significant advances in the synthesis of SWCTs. This will allow us to obtain long, defect free and therefore strong nanotubes with the strength much higher than that of steel, as theoretically predicted.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	As a result of this program it is found that the introduction of a rotating cathode as a first method to simulate microgravity conditions in arc-discharge synthesis of single wall carbon nanotubes (SWCNT) alters the plasma environment during arc discharge and produces carbon nanotubes with various unique morphologies. Rotating Cathode System (RCS) has increased overall yield produced during synthesis, produced large diameter SWCNT bundles, and produced nanotube bundles with bimodal diameters distribution as evidenced by Radical Breathing Modes (RBMs) Raman Spectra. Further experimentation should enable more control over morphology and diameters and reveal further insight into carbon nanotube growth mechanisms in simulated microgravity. Arc discharge in a multielectrode geometry: second method for simulating the microgravity conditions is a viable production technique and an important experimentation tool for discovering the growth mechanisms of carbon nanotubes. It allowed us to obtain SWCNTs of different distributions and outcome by changing the relative geometries of two cathodes, relative to a central anode. The CVD methods have been developed, which allow to create MWCNTs of spiral and wavy geometries, which are particularly interesting for composites reinforcement. It has been discovered that the forests of MWCNTs created by the CVD can be transformed from one electrode to another, and this invention is filed as a US patent. The composites of synthesized SWCNTs with conjugated polymers have been created, and their thermal and mechanical properties are being studied.
Bibliography Type:	Description: (Last Updated: 06/01/2006)
Articles in Peer-reviewed Journals	Zhang M, Fang S, Zakhidov AA, Lee SB, Aliev AE, Williams CD, Atkinson KR, Baughman RH. "Strong, transparent, multifunctional, carbon nanotube sheets." Science. 2005 Aug 19;309(5738):1215-9. PMID: 16109875 , Aug-2005
Articles in Peer-reviewed Journals	Al. A. Zakhidov, R. Nanjundaswamy, M. Zhang, S.B. Lee, W.M. Sampson, A.N. Obraztsov, A. Cunningham, A.A. Zakhidov. "Spark Light Emission Coupled with the Field Emisson from Carbon Nanotubes." Journal of Applied Physics (in press), 2006. , Jun-2006
Articles in Peer-reviewed Journals	R. Ulbricht, X. Jiang, S. Lee, M. Zhang, S. Fang, R. Baughman, A. Zakhidov. "Polymeric Solar Cells with Oriented and Strong Transparent Carabon Nanotube Anode. " Physica Status Solidi (in press), 2006. , Jun-2006